

XRPL Validator — Daily Engineering Report • 2026-07-17

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **14 commits** (xrpl-validator 14)
- 2 test / vector commits

Engineering changes

- **feat(node): parity_probe bin — runtime fixture replay for the scout loop** (52e9df5 , xrpl-validator) Loads blobs+expected fixture files by path (no rebuild per ledger) and reports the standard parity block. Exit 0 clean / 1 divergent / 2 fixture error.
- **feat(node): network-parameterized replay — NetParams for testnet/devnet parity probes** (8ed5545 , xrpl-validator) apply_ledger_in_order_with_net carries network_id/base_fee/reserves (mainnet wrapper keeps every existing caller unchanged); fixtures capture the params from server_state; parity_probe replays under the fixture's network rules.
- **feat(node): differential_probe — measure native engine vs mainnet parity** (c862c37 , xrpl-validator) Replays each fixture tx through the native xrpl_ledger transactors and compares (TER, mutation-key-set) to mainnet — the instrument for the fully-Rust roadmap. Fixture generator now emits tx_json + tx_order for the native leg.
- **fix(node): differential_probe --json emits only JSON (SUMMARY to stderr); re-mint sample fixtures with tx_json** (9cfb9af , xrpl-validator)
- **fix(ledger): TrustSet decodes base58 issuer (was hex-only → every real tx temMALFORMED)** (fdca115 , xrpl-validator) extract_limit_amount decoded the LimitAmount issuer with hex::decode, but mainnet JSON carries it as a base58 classic address — so every real TrustSet failed to Malformed before applying. Add decode_issuer (hex OR base58 via the canonical xrpl_core AccountId::from_address). Differential: TrustSet DIVERGE-TER 16->0 (all now tesSUCCESS matching mainnet, 1 full MATCH; the remaining 15 are DIVERGE-MUT — owner-directory ...
- **feat(node): differential_probe loads pre-state objects (not just AccountRoots)** (330ed4d , xrpl-validator) Native was starved of pre-state — only AccountRoots were loaded, so it phantom-created every RippleState/Offer/Check/AMM object it referenced, inflating the whole conformance map. Now loads each tx's Modified/Deleted affected-nodes at seq-1 (transcoded to native's account-hex convention), so native SEES the objects it operates on. TrustSet lines now modify instead of phantom-create; the residual misses are genuine ...
- **fix(ledger): NFTTokenAcceptOffer reads offer ref as hex ledger-index, not {Account,Sequence}** (31735dc , xrpl-validator) NFTTokenSellOffer/BuyOffer is the offer object's 32-byte hex ledger index; the native code treated it as an {Account,Sequence} pair and failed to Malformed before ever reading the offer. Now parses the hex index and takes the owner from the loaded offer object. Genuine bug fix; the transactor still has further naive parses (NFTokenID-as-object) + needs NFT-page transfer logic to green — noted, not chased here. ...

- **feat(ledger): owner-directory bookkeeping (single root page) + wire into TrustSet** (274d420 , xrpl-validator)
New crate::ledger::directory: owner_dir_insert / owner_dir_remove maintain an account's owner directory (the DirectoryNode listing objects it owns). Shared infrastructure for every object-creating transactor. TrustSet now inserts a new RippleState into BOTH parties' directories (owner_dir_insert), so native emits the DirectoryNode mutations mainnet does — at the right key + kind (the gate compares key/kind sets, so ...
- **feat: multi-page owner directory + load native-read keys — TrustSet 1->10 MATCH** (a22af92 , xrpl-validator) Two increments that together get trust-line creates to byte-for-key parity: - directory.rs: owner_dir_insert now walks the page chain (root.IndexPrevious -> last page), appending to the last page or splitting to a new dir_page_key page on overflow — so native touches the SAME directory page mainnet does (large gateway issuers keep trust lines on non-root pages). - differential_probe: native_read_keys loads the ...
- **feat(xrpl-ledger): trust-line deletion + oracle state threading — TrustSet 14/14 MATCH** (87d6ff7 , xrpl-validator)
- trust_set: rippled-faithful default-line deletion (per-side interest test: limit/balance/quality/freeze/auth/NoRipple-vs-DefaultRipple), tf-flag application (tfSet/ClearNoRipple, tfSet/ClearFreeze) before the test, creation flags (creator reserve bit + auto-NoRipple), both-roots OwnerCount touch at delete mirroring create - directory: hint-based owner_dir_remove (rippled's no-walk LowNode/HighNode design) with ...
- **feat(xrpl-ledger): OfferCancel directory unlinking — 6/6 MATCH across two ledgers** (270345a , xrpl-validator)
- directory: generalize dir_remove(root_key, ...) so order-book quality dirs share the owner-dir page mechanics; owner_dir_remove is now a wrapper. Hint path no longer requires the root page loaded (book hints often ARE the root). Empty-root cascade: deleting the only non-root page kills the whole directory when the root holds no entries (rippled behavior). - directory: hex-case normalization — mainnet-loaded ...

Tests and campaigns

- **test(node): arm scout find #105663160 — NFTokenAcceptOffer silent-TER parity gate** (3bac75a , xrpl-validator)
- **test(node): 5 coverage fixtures (AMM/Check/NFToken-accept) for engine-wide conformance map** (f70baf9 , xrpl-validator)

Docs and operations

- **revert NFToken offer-ref parse (needs focused pass); keep AMM XRP+base58 encoding fix** (51c9665 , xrpl-validator)

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Validator Bonding With Slashing-Backed Finality** (consensus) Add a new owner-reserved SLE `ltVALIDATOR_BOND` (`keylet::validatorBond(nodePublicKey)`) holding `sfAccount`, `sfValidatorPublicKey`, `sfBondAmount` (escrowed)
- **Concentrated Liquidity Ranges for AMM Pools** (amm-defi) Add a new owner-reserved SLE `ltAMM_POSITION` (`keylet::ammPosition(ammID, account, sequence)`) carrying `sfTickLower`, `sfTickUpper` (fixed-point ticks on
- **Domain-Gated Eligibility for MPT Transfers** (rwa) Add `sfDomainID` (the existing `PermissionedDomain` ID type already used by `PermissionedDEX`) to `MPTokenIssuance`, settable via a new `tfRequireDomain` fl

Appendix — commits

Time	Repo	Commit	Subject
11:07	xrpl-validator	52e9df5	feat(node): parity_probe bin — runtime fixture replay for the scout loop
11:57	xrpl-validator	3bac75a	test(node): arm scout find #105663160 — NFTokenAcceptOffer silent-TER parity gate
13:04	xrpl-validator	8ed5545	feat(node): network-parameterized replay — NetParams for testnet/devnet parity probes
15:34	xrpl-validator	c862c37	feat(node): differential_probe — measure native engine vs mainnet parity
15:40	xrpl-validator	9cfb9af	fix(node): differential_probe --json emits only JSON (SUMMARY to stderr); re-mint sample fixtures with tx_json
16:16	xrpl-validator	f70baf9	test(node): 5 coverage fixtures (AMM/Check/NFToken-accept) for engine-wide conformance map
16:27	xrpl-validator	fdca115	fix(ledger): TrustSet decodes base58 issuer (was hex-only → every real tx temMALFORMED)
16:43	xrpl-validator	330ed4d	feat(node): differential_probe loads pre-state objects (not just AccountRoots)
16:47	xrpl-validator	31735dc	fix(ledger): NFTokenAcceptOffer reads offer ref as hex ledger-index, not {Account,Sequence}
17:11	xrpl-validator	51c9665	revert NFToken offer-ref parse (needs focused pass); keep AMM XRP+base58 encoding fix
17:29	xrpl-validator	274d420	feat(ledger): owner-directory bookkeeping (single root page) + wire into TrustSet
17:43	xrpl-validator	a22af92	feat: multi-page owner directory + load native-read keys — TrustSet 1->10 MATCH
19:20	xrpl-validator	87d6ff7	feat(xrpl-ledger): trust-line deletion + oracle state threading — TrustSet 14/14 MATCH
19:50	xrpl-validator	270345a	feat(xrpl-ledger): OfferCancel directory unlinking — 6/6 MATCH across two ledgers